Arkanil Paul

arkanil.paul.work@gmail.com | Website | LinkedIn | Github

Research Interests

My research interests lie in computer architecture and systems, with a focus on cache and memory systems, on-chip interconnects (including 3D NoCs), CPU/GPU microarchitecture, ML/DL accelerators, and high-performance computing. I work across the stack, from RTL design in SystemVerilog and architectural simulation in gem5, ONNXim and GPGPU-Sim to workload profiling and systems software in C/C++.

Education

Indian Institute of Technology, Roorkee Roorkee, India
Integrated M.Tech. in Geophysical Technology; CGPA: 8.054/10 Jun 2026

Thesis: Integrating Geophysical Analysis and Machine Learning for Landslide Spatial Probability Mapping in the Garhwal Himalaya, India

Relevant Coursework

Computer Architecture and Microprocessors (CSN-221); Computer Architecture for AI (DA-408);
Discrete Mathematics (MAC-102); Fundamentals of Electronics (ECN-102); Graph Theory (MA-416)

Research Experiences

School of Electrical and Computer Sciences, IIT Bhubaneswar Bhubaneswar, India

- *Project Associate, NPU Simulation and GPU Microarchitecture* 08/2026 - Present
  - Working on NPU simulation with ONNXim to integrate 3D NoC interconnects through reusable interfaces.
  - Beginning research on Blackwell RTX 5000/6000 Pro GPUs, focusing on cache policies and memory behavior.

Department of Earth Sciences, IIT Roorkee Roorkee, India

- *Dissertation, Landslide Spatial Probability Mapping (advised by Prof. Pallavi Chattopadhyay)* 06/2025 - 05/2026
  - Prepared 5,621 labelled samples and 32 model features from terrain, rainfall, land-cover and geological inputs.
  - Selected XGBoost over MLP and Random Forest using Optuna to tune hyperparameters and 5-fold CV to compare models, attaining 0.92 test ROC-AUC and 87.7% landslide recall for mapping regional landslide susceptibility.
  - Built a 4-stage Python pipeline to align rasters, encode pixel features, refit the selected model and predict landslide susceptibility across 3.82M pixels, exporting georeferenced GeoTIFF maps with explicit NoData handling.
  - Interpreted ERT and VLF-EM surveys at 3 landslide sites for qualitative corroboration of mapped susceptibility.

Department of Computer Sciences and Engineering, IIT Delhi New Delhi, India

- *Research Intern, Cache Bandwidth Management* 05/2025 - 08/2025
  - Implemented LLC bandwidth management policies for multiple cores and accelerators in gem5 source code.
  - Debugged segmentation faults & simulation failures in gem5, reducing crashes & improving simulation stability.
  - Evaluated policy performance on 6 CPU cores and 4 accelerators by conducting SPEC and microbenchmarks.
  - Identified bottlenecks & optimizations analysing microarchitectural metrics like IPC, miss-rate, latency, etc.

School of Electrical and Computer Sciences, IIT Bhubaneswar Remote

- *Research Intern, Federated Learning and Microarchitectural Analysis* 09/2024 - 04/2025
  - Developed an asynchronous Federated Learning framework in C++ (5k LOC) using Boost and LibTorch, with a plug-in architecture featuring concurrent clients, on-demand model integration & lazy loading large datasets.
  - Applied staleness-aware FedAsync aggregation to incorporate client updates without waiting for slower clients.
  - Profiled 5+ server and client systems using AMD uProf and contributed to detailed microarchitectural analysis.
  - Validated framework scalability using load testing with 150 parallel clients (50 per model) across 3 DL models.
- *Research Intern, CPU/GPU Workload Profiling* 03/2024 - 07/2024
  - Containerized simulation environments with Docker, reducing setup time, and eliminating configuration errors.
  - Executed GPGPU-Sim experiments on 7 DL models to support workload characterization & bottleneck analysis.
  - Generated 200GB+ of detailed execution traces with Intel Pin tool for complex workload analysis on ChampSim.

Projects

Mehta Family School of Data Science and Artificial Intelligence, IIT Roorkee Roorkee, India

- *ANN Accelerator on FPGA (Individual project, coursework DA-408)* 09/2025 - 10/2025
  - Designed configurable neural-network RTL (MNIST Digit Classifier) in SystemVerilog with 8 MAC units.
  - Integrated UART transmit and receive state machines for 8-bit grayscale image input and serial digit output.
  - Verified neural-network inference on a physical Basys 3 FPGA, attaining 95% accuracy on 20 project test images.
  - Met 100 MHz timing constraints in Vivado with 0.445 ns setup slack, using 34 block RAM tiles and 32 DSPs.

Self-directed

- *AsyncTiny (Individual Project)* 10/2023 - 01/2024
  - Developed a C++11 (no STL) async runtime library (3k LOC) featuring coroutines, futures, tasks, timers.
  - Supports cooperative scheduling, conditional sleep, task cancellation, & runs on Arduino, ESP32, Linux, etc.
  - Achieves ∼200B RAM overhead for 20 concurrent tasks via zero-copy task reuse & memory efficient design.

Department of Electronics and Communication Engineering, IIT Roorkee Roorkee, India

- *Transformers (Team project, coursework ECN-102)* 03/2023 - 05/2023
  - Designed a humanoid-to-car robot under Arduino control, using 14 servos and 4 DC motors for actuation.
  - Engineered power and control circuits for lithium-ion batteries, BMS, voltage converters and motor actuation.
  - Programmed C++ control abstractions for 2 motor types, joints and wheels, with a Bluetooth remote interface.
  - Demonstrated functional correctness of wired electronics for the 18-motors robot design under Arduino control.

Self-directed

- *MazeRush (Team Project)* 01/2023 - 02/2023
  - Created a top-down 2D maze/action game with 6 levels, each introducing a new map and entity for varied play.
  - Structured gameplay using OOP concepts and factory classes to reuse common behavior & spawn 6 enemy types.
  - Added BFS pathfinding and raycast line-of-sight detection for enemy navigation and pursuit across 2D tilemaps.
  - Produced a Windows desktop build in Unity/C#, bundling 2D scenes, scripts and assets into a playable game.

Department of Computer Science and Engineering, IIT Roorkee Roorkee, India

- *Configurable Set-associative Cache Simulator (Team project, coursework CSN-221)* 09/2022 - 10/2022
  - Created a set-associative cache simulator in SystemVerilog with LRU, write-back and write-allocate policies.
  - Achieved 95% hit rate through validation on LU decomposition, matrix multiply (32x32) & quicksort workloads.
- *5-stage Pipelined Processor (Team project, coursework CSN-221)* 07/2022 - 09/2022
  - Implemented inst. fetch/decode, control, ALU, memory & branch mechanism in SystemVerilog, with testbenches.
  - Created MIPS-like custom ISA (consisting of 80 different instructions) and standalone assembler in modern C++.
  - Demonstrated Fibonacci (up to 47th number) and Factorial (up to 12!) MIPS instructions with minimal stalls.

Achievements

Global Rank 1005 out of 33,953 participants in ICPC 2026 Online Challenge 1 powered by Huawei	Aug 2026
Global Rank 30 out of 18,557 participants in Codechef May Long One 2022 (Division 3)	May 2022
Global Rank 26 out of 19,211 participants in Codechef April Long Two 2022 (Division 3)	Apr 2022
JEE Advanced: All India Rank 7481 out of 141,699 candidates	2021
JEE Mains: All India Rank 9811 out of 1,019,133 candidates (99.037 percentile)	2021

Technical Skills

Programming: C, Modern C++, C#, Python, SQL, Verilog/SystemVerilog

Libraries: C++ STL, Boost, LibTorch, Python libraries (XGBoost, Optuna)

Tools: AMD uProf, Intel Pin, gem5, ONNXim, GPGPU-Sim, ChampSim, Vivado Design Suite, Docker, Git, Linux

Hardware: FPGA (Basys 3), Arduino, ESP32

Positions of Responsibility

Programming and Algorithms Group, IIT Roorkee Roorkee, India

- *Member* 04/2022 - 07/2022
  - Hosted open lectures on algorithms and programming fundamentals, engaging over 200 attendees on campus.